UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 006 · Problems on Euler's theorem · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S006·12	Deck 006, slide 12 (rendered .pptx)
T006a 00:05:18	Transcript 006 part I (slides 3–4, 9.1 min)
T006b 00:08:20	Transcript 006 part II (slides 5–17, 12.9 min)
CW U1 P20	Unit 1 class-work problem 20
ESA Jan-24 s3 1c	Past paper: Jan 2024, set 3, question 1.c

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6. Part B · Symbol glossary

A Formula sheet

1 Testing homogeneity: the scaling test

E1.1 · Scaling test ESA ★★

$$u(\lambda x, \lambda y) = \lambda^n\, u(x,y) \implies u \text{ homogeneous of degree } n$$

GIVES	Whether u is homogeneous, and its degree.
APPLIES	Any function; with three variables scale z too.
NOTE	If no power λ^n factors out, u is not homogeneous.

S006·5 · T006b 00:00:52

E1.2 · Degree of a quotient ESA ★★

$$\deg \frac{P}{Q} = \deg P - \deg Q$$

GIVES The degree without writing out λ.

APPLIES P, Q homogeneous.

NOTE $\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$: $1 - 4 = -3$.

ADDED FOR CLARITY S006·4 · T006a 00:06:10

2 Verifying Euler's theorem directly

E2.1 · Euler's theorem ESA ★★★

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\,u$$

GIVES The first-order Euler sum.

APPLIES u homogeneous of degree n; add $z\,u_z$ for three variables.

NOTE To verify, compute the left side by differentiating and show it equals nu.

S006·3 · S006·10 · T006a 00:00:00

3 When u itself is homogeneous: first and second order

E3.1 · Second order ESA ★★★

$$x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = n(n-1)\,u$$

GIVES The second-order Euler sum.

APPLIES u homogeneous of degree n.

NOTE $n = -1$ gives $2u$; $n = -\frac{2}{3}$ gives $\frac{10}{9}u$.

S006·11 · T006b 00:07:23

E3.2 · First plus second order ESA ★★★

$$x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} + x\,u_x + y\,u_y = n^2u$$

GIVES The combination asked in ESA Jan 2024 sets 3 and 4.

APPLIES u homogeneous of degree n; for $u_1 + u_2$ use $n_1^2u_1 + n_2^2u_2$.

NOTE $n(n-1) + n = n^2$.

ADDED FOR CLARITY ESA Jan-24 s3 1c · ESA Jan-24 s4 1d · CW U1 P23

4 When f(u) is homogeneous: the extension

E4.1 · Extension, first order

ESA ★★★

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=n\frac{f(u)}{f'(u)}=g(u)$$

GIVES The Euler sum when u is not homogeneous but $f(u)$ is.

APPLIES $f(u)$ homogeneous of degree n; add $z\,u_z$ for three variables.

NOTE $\sin\to n\tan u$, $\tan\to\frac{n}{2}\sin 2u$, $e^u\to n$, $\log\to nu\log u$.

S006·13 · S006·15 · S006·17 · T006b 00:09:08

E4.2 · Extension, second order

ESA ★★★

$$x^2u_{xx}+2xy\,u_{xy}+y^2u_{yy}=g(u)\left[g'(u)-1\right]$$

GIVES The second-order sum for the extension.

APPLIES g from E4.1; differentiate g with respect to u.

NOTE If g is a constant (the $\log$ case), the answer is $-g$.

S006·15 · S006·17 · T006b 00:10:59

E4.3 · Deriving the extension by the chain rule

ESA ★★

$$F=f(u):\ xF_x+yF_y=nF,\quad F_x=f'(u)\,u_x\ \Rightarrow\ f'(u)\,(xu_x+yu_y)=nf(u)$$

GIVES Where E4.1 comes from; T006a works slide 006·4 this way.

APPLIES When you prefer not to quote E4.1.

NOTE On slide 006·4: $\cos u\,(xu_x+yu_y+zu_z)=-3\sin u$; divide by $\cos u$.

S006·4 · T006a 00:07:33 · T006a 00:08:22

B Symbol glossary

Symbol	Name	Meaning	Appears in
λ	Scale factor	Replaces x, y by $\lambda x, \lambda y$ in the scaling test	E1.1
n	Degree	Of u (topic 3) or of $f(u)$ (topic 4)	E1.1–E4.2
$f(u)$	Homogeneous function of u	$\sin u$, $\tan u$, e^u, $\log u$	E4.1
$f(y/x)$ (slide 3)	Arbitrary function of the ratio	$Z = \sqrt{x}\, f(y/x)$; not the $f(u)$ of topic 4	W2.1
$g(u)$	Extension function	$nf(u)/f'(u)$; equals $xu_x + yu_y$	E4.1, E4.2
$g'(u)$	Derivative of g	With respect to u	E4.2
Z	Auxiliary function	$\cos u$ (slide 3) or $\sin u$ (slide 4, transcript)	W2.1, W4.1
z	Third variable	Lowercase, on slide 4 only	W4.1
$u_x,\ u_{xy}$	Partial derivatives	$\partial u/\partial x$, $\partial^2 u/\partial x \partial y$	E2.1, E3.1
u_1, u_2	Homogeneous parts	Parts of different degrees in a sum	E3.2

Remark. Two clashes come from the slides themselves. On slide 006·4 lowercase z is the third variable while the lecture writes the auxiliary function as uppercase $Z = \sin u$; on slide 006·3 $Z = \cos u$ is the auxiliary function. And f is both the homogeneous function of u ($f(u) = \sin u$, $\log u$, e^u) and, on slide 006·3, the arbitrary function in $\sqrt{x}\, f(y/x)$.